

Reduce the Leakage Current and Power in VLSI Technology

Nisha Sharma^{1*}, Laxmi Singh², Shilpa Tripathi³

Abstract

Demand for portability Reduce and increase in size, low cost, low dynamic power, low leakage current, low sub threshold current, and the necessity to minimise at least one of these. Circuit complexity increases with circuit integration. Memory and microprocessors are included in mass-produced chips. Microcomputer systems' central system memory uses dynamic random-access memory (DRAM). One benefit of DRAM memory is its high speed and capacity integration. The examination of DRAM memories with one and three transistors is presented in this research article. Three transistor DRAM implementations have the benefit of dynamic power dissipation reduction and recurrent precharge cycles. The quantity of DRAM activity, such as the possibility of random read/write operations, compact cell size in contrast to SRAM, incredibly unified, extremely low cost of production Often utilised as primary memory, Data stored is erratic; a cyclic refresh is required; speed is medium. DRAM is essentially a memory array with individual bit access; it can read and write data. These research articles also cover the DRAM's read, write, and hold capabilities as well as charge leakage and how it happens. To design and implement 1T and 3T DRAM, we employ Cadence Virtuoso Tool Design Systems, which simulates in an analogue design environment. The analysis of reducing low dynamic power is shown here.

Keywords: High performance, low cost, leakage current, low power

INTRODUCTION

We concentrate on the DRAM cell in this research. System-on-a-Chip (SOC) and personal computer systems use DRAM architectures. Here, we examine and assess a DRAM cell's effective gate leakage current during READ, WRITE, and REFRESH operations. The study and mitigation of leakage current have been difficult in sequential and combinatorial circuits [1]. If leakage current is present, memory circuit design needs to be given additional consideration. This paper illustrates the DRAM's pre-charge cycle as well as its read and write data operations. Workstations and personal computers are the primary uses for DRAM. Memory is essentially a network of electrically charged locations where computers

store data in the form of "0s" and "1s" that are easily accessed. The PC processor's random-access feature allows it to access any location in memory directly [2-4]. DRAM is dynamic because, in contrast to static RAM (SRAM), it needs a fresh electrical charge every few milliseconds to maintain the freshness of its storage cells. DRAM has a far higher capacity per unit of surface area than SRAM since it is far less expensive per storage cell and each storage cell is relatively simple.

ANALYSIS OF A DRAM OPERATION

Dynamic random-access memory (DRAM) is a kind of random-access memory where each piece of data is stored inside an integrated circuit in its own

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Received Date: January 10, 2024

Accepted Date: January 11, 2024

Published Date: January 25, 2024

Citation: Nisha Sharma, Laxmi Singh, Shilpa Tripathi. Reduce the Leakage Current and Power in VLSI Technology. Journal of VLSI Design Tools & Technology. 2023; 13(3): 19–25p.

capacitor. There are two states for the capacitor: charged and discharged. It is thought that these two states correspond to the conventional bit values of "0" and "1". Because capacitors leak charge, information eventually ages if the charge is not regularly replaced. Because of this refresh requirement [5-8], it is a dynamic memory as opposed to SRAM and other static memory. In personal computers, dynamic random-access memory (or "RAM") functions as the main memory (DRAM). It is used in workstations and laptops in addition to video games.

A transistor plus a capacitor completes a single-transistor (1T1C) DRAM cell, as shown in Figure 1.

RAM data is temporarily stored on the storage capacitor via leakage currents that reduce charge. Charge capacity is subject on several conditions. 1. The charge capacity is 1. If C_s is charged to V_s .

$$Q_s = C_s V_s \quad (1)$$

2. If $V_s = 0$ and $Q_s = 0$, then the logic is 0.

3. If $V_s = \text{big}$ and $Q_s > 0$, then the logic is 1.

Write Operation in a DRAM Cell

The access transistor is turned on and set to $WL = V_{DD}$ for the write operation. In the event where $V_d = 0$, $V_s = 0$, $Q_s = 0$, and the threshold voltage for n $V_{tn} = V_{DD}$, logic 1 is stored if $Q_s = C_s(V_{DD} - V_{tn})$. When $V_s = V_{DD}$ and $V_d = V_{DD}$, logic 0 is reached and $Q_s = C_s(V_{DD} - V_{tn})$.

Charge on the capacitor

$$Q_s = C_s V_s \quad (2)$$

For writing 1, Figure 1 illustrates writing operations in DRAM.

$$V_s = V_{max} = V_{DD} - V_{tn} \quad (3)$$

$$Q_{max} = C_s(V_{DD} - V_{tn}) \quad (4)$$

Read operation in a DRAM Cell

When the access transistor is turned on for reading, the charge on C_s is transferred to C_{bit} , the bit line capacitance. Read Operations in DRAM shown in Figure 2. This causes a fluctuation in the bit line voltage, V_{bit} , which is amplified to read a 1 or 0. First charge on C_s is redistributed as $Q_s = C_s V_s$, and C_{bit} is subsequently redistributed till $V_{bit} = V_s = V_f$ (final voltage).

$$Q = C_s + C_{bit} + V_f \quad (5)$$

Because of the conservation of charge

$$C_s V_s = V_f(C_s + C_{bit}) \quad (6)$$

Always less than V_s

$$V_f = C_s V_s / (C_s + C_{bit}) \quad (7)$$

Because charge dispersal erases the data that has been stored in DRAM, read operations must include simultaneous rewriting. This makes DRAM read operations crucial. Bit lines are pre-charged to $V_{DD}/2$, the selected word line is enabled, and the capacitance of the input and output signal share charge.

$$Q_s = C_s V_s \quad (8)$$

$$Q_s = C_s V_s + C_{bit} V_f \quad (9)$$

$$V_f = \left(\frac{C_s}{C_s + C_{bit}} \right) V_s \quad (10)$$

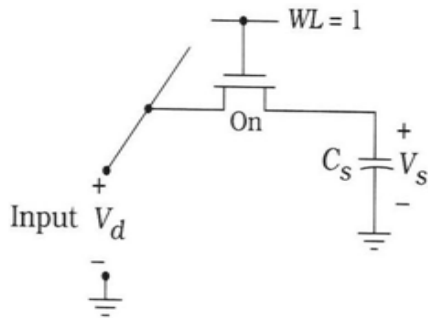


Figure 1. Write operations in a DRAM.

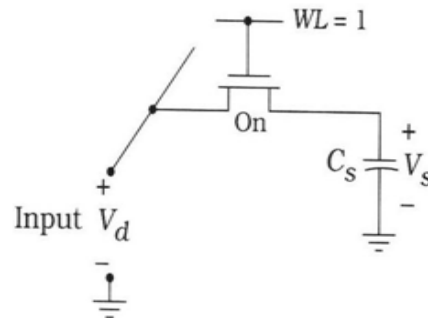


Figure 2. Read operations in a DRAM.

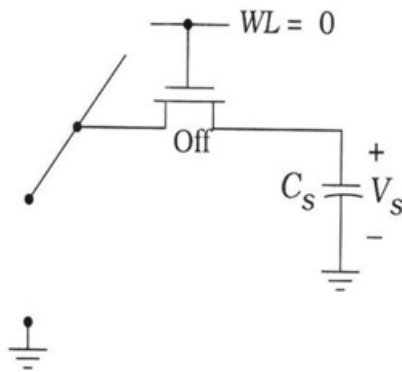


Figure 3. Hold operation in a DRAM cell.

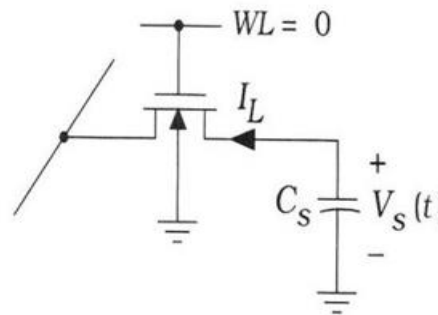


Figure 4. Charge leakage in a DRAM cell.

Hold operation in a DRAM Cell

The access control transistor is off during hold operation, $WL = 0$, and the charge is held on C_s . Because the access transistor leaks when it is turned off, leakage currents slowly discharge C_s during Hold. If I_L is known, the discharge time may be computed using the formula $I_L = -\delta Q_s / \delta t = -C_s \delta V_s / \delta t$. The longest period that the voltage on C_s is high enough to result in a logic 1 is known as the hold time as shown in Figure 3. $V_{max} = V_1$ is the amount of discharge time. The needed lengthy hold time can be obtained by solving the equation $t_h = (C_s / I_L)(\Delta V_s)$, assuming that I_L is constant. Moreover, T_h grows when C_s and I_L change; a typical value is $t_h = 40 \mu\text{sec}$, where $I_L = 1 \text{ nA}$, $C_s = 40 \text{ fF}$, and $\Delta V_s = 1 \text{ V}$.

Charge Leakage

When $V_G < V_T$, a MOSFET is turned off but still allows very small leakage currents. Leakage current: i_{L1} pn-junction current that leaks to substrate and i_{L2} below threshold power to bit line as shown in Figure 4.

Data must be retained under worst Conditions when Maximum junction temperature and Lowest bit line voltage.

Refresh Rate

A DRAM controller's refresh logic handles the periodic refresh automatically; no additional hardware or software is needed. Data is only stored momentarily in DRAM, which is "dynamic." Refresh Data must be read, rewritten, amplified, and refreshed on a regular basis for the operation to retain data for as long as power is applied. The frequency at which cells in a DRAM circuit must be updated to maintain data is known as the refresh rate. Refresh is equal to $1/2$. Restart.

Refresh Operation

There's a cyclic refresh because of information deterioration caused by leakage current. Word line activation and bit line recharge Information sensing and restoration are necessary for read operations. Multiple rows can be refreshed at once, and the specs specify the maximum refresh time of 64 ms per cell. The amount of power consumed and the resulting current limit the number of rows. The one transistor DRAM with capacitor is now displayed, and Figure 5's word line is used to control the circuit. The 3T DRAM with pre-charge evaluation signal is now also visible. All of the peripheral circuits of the 3T DRAM cell operate on a two-phase, non-overlapping clock system.

Figure 6 illustrates a simulation of a 1T1R1C DRAM, showing how the capacitor charges during a write operation.

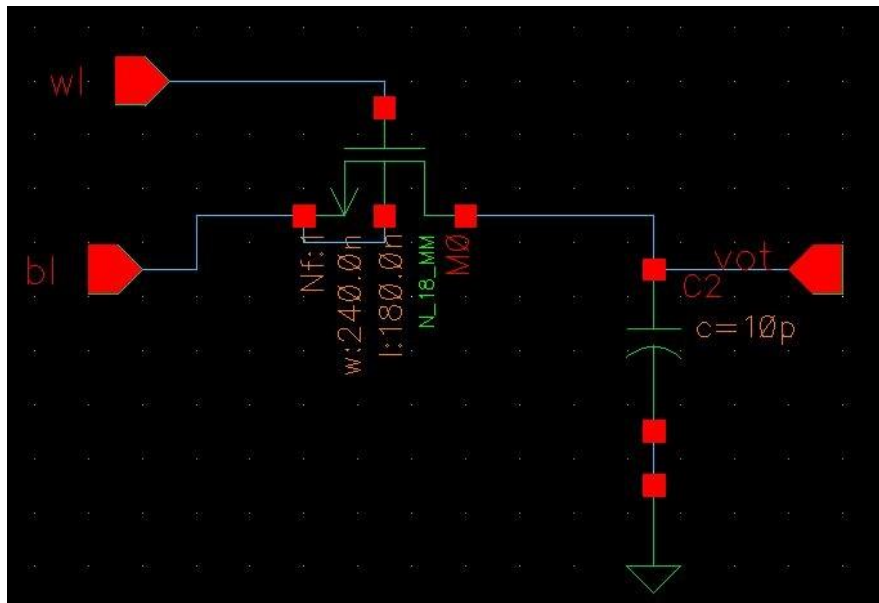


Figure 5. Schematic of a 1T DRAM.

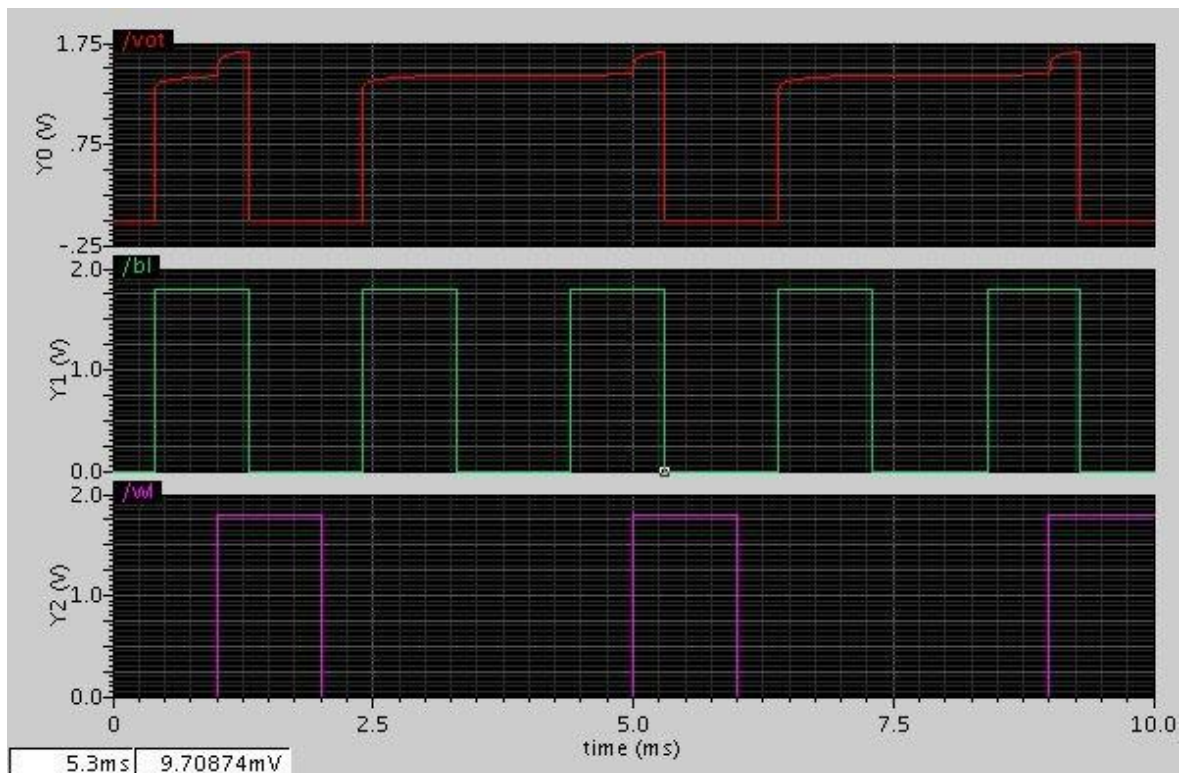


Figure 6. Waveform of 1T DRAM.

While ϕ_1 drives the pre charge actions, ϕ_2 determines the read and write actions. Before every data read and write operation, pre charge cycles are carried out. These cycles begin when the pre charge signal PC reaches high. Throughout the pre-charge cycle, column pull-up transistors are ON and the corresponding column capacitances, c_2 and c_3 , are charged to logic high levels. Every single data read and write operation is completed when the PC is low, or during the active ϕ_2 phase. When doing four consecutive operations on a 3T DRAM cell, the common voltage waveform is written 1, read 1, write 0, and read 0. The transient current that charged the two columns during the pre-charge cycle essentially ends the PC when both capacitance voltages reached their steady state values. when the two column capacitances (c_2 and c_3) are at least one order of magnitude smaller than the internal store capacitance. Because the data that needs to be written on the DRAM cell is a logic "1," the inverted data input for write "1" operations is at the logic low level. Consequently, the data write transistor is turned off. [9-10] Once the write '1' operation is complete, the write access transistor m_1 is turned off, and the store capacitance charge prevents the logic high level transistor m_2 from conducting. When the read access transistor m_3 goes on, the data read column capacitance c_3 and the ground are connected by a conducting path.

This preserves the charge held in c_1 and permits multiple reads of the 3T DRAM cell as shown in Figure 7. Since the data that needs to be written on the DRAM cell is logic "0," write "0" operations require inverted data input that is at the logic high level. The data write transistor is therefore turned on. As the read access transistors m_3 turns on, there isn't a conducting connection whatsoever between the data read column capacitance c_3 and the ground.

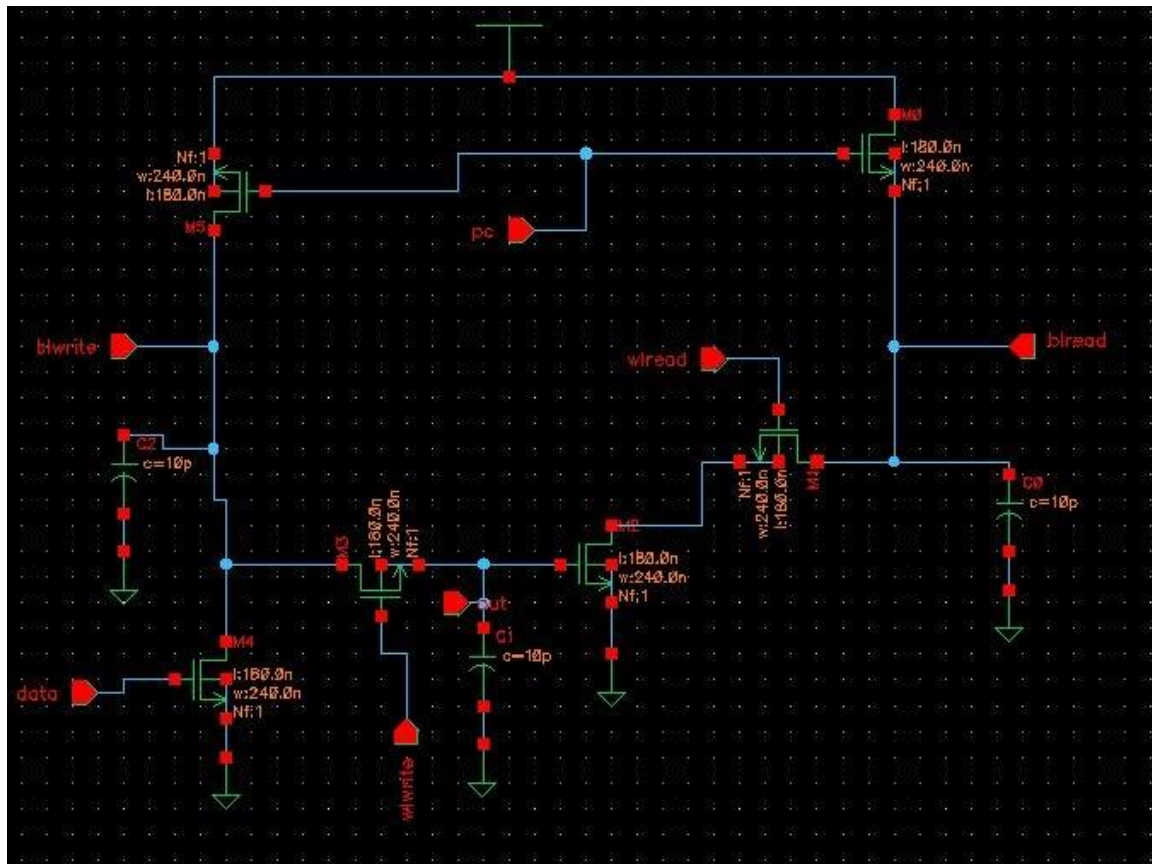


Figure 7. Schematic of a 3T DRAM with Pre charge cycle.

EXPERIMENTAL RESULTS

To design and construct circuits, utilise Cadence Design Systems, which simulates an analogue design environment. The current for the 1T DRAM circuit is 4.3 mA, and the current for the 3T DRAM circuit is 8.3 mA when we examine the circuit with $V_{DD} = 1.8V$. One advantage of DRAM is its structural simplicity: each bit only requires one transistor and one capacitor. This allows DRAM to achieve very high densities. DRAM is volatile memory. Because when the power is turned off, the data is soon lost and can be recovered by regularly refreshing. A single memory chip can contain several billions of the tiny transistors and capacitors that are required.

Figures 8 & 9 shows the waveform of 3T DRAM pre charge cycle with read and write operation.

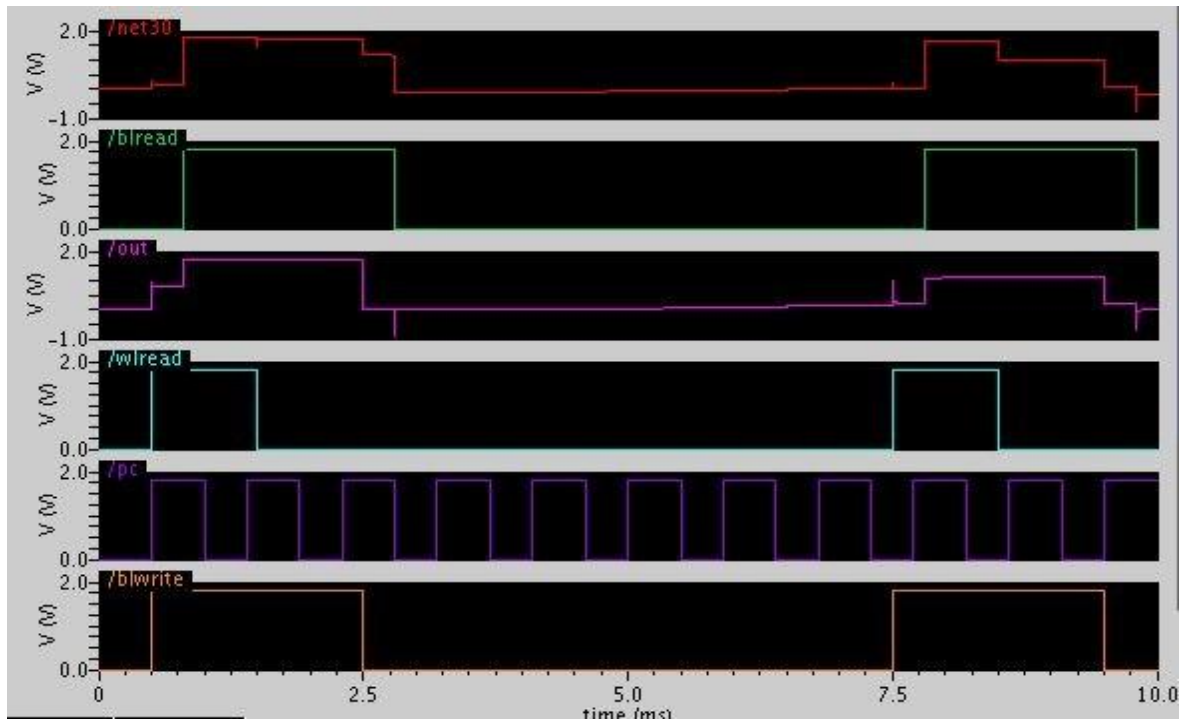


Figure 8. Waveform for read operation.

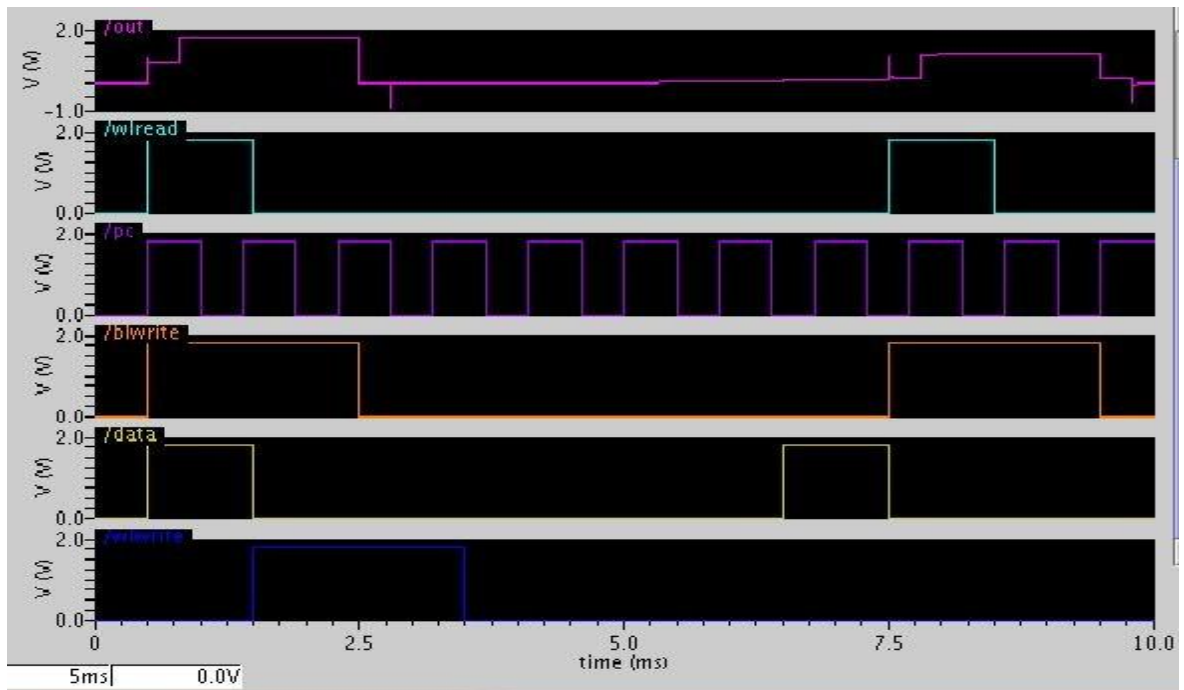


Figure 9. Waveform for write operation.

CONCLUSIONS

The 3T dynamic RAM cell under investigation here does not supply any static power for data storage; instead, it operates in one of the several DRAM operation modes that are covered here, where there is no continuous current flowing through the circuit. Additionally, modelling recommends routinely utilising a pre-charge cycle to reduce dynamic power loss. By applying leakage current reduction, various strategies such as power gating for low dynamic power or multithreshold voltage for leaking power can be used to reduce leakage current and power dissipation.

Acknowledgment

The authors would like to thank Dr. Laxmi Singh for their helpful support.

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